

Estimating the Unrecorded Game

Coverage, Bayesian Models, and Explicit Imputation for Historical Baseball Play-by-Play

September 13, 2026

Abstract

Historical baseball play-by-play records outcomes much more consistently than pitch sequences, batted-ball geometry, fielding attribution, or game conditions. We distinguish the process that generates a game from the process that records it, then combine hierarchical models, deterministic constraints, and explicitly labelled empirical fallbacks over 18,141,020 events in 205,886 acquired play-by-play games from 1903 through 2025. The imputation candidate maps 116 target fields to sixteen additive database surfaces, including games before 1989; games known only through box scores or season totals are outside its scope.

The statistical contribution is a treatment of selective observation and its limits. Deterministically recoverable ground balls constrain the missing trajectory distribution from below, but do not identify a missing-not-at-random selection offset. Historical model experiments illustrate this distinction; their older validation labels do not establish current calibration or historical transport. Under the current evidence contract, all 24 retained legacy artifact pointers are explicitly exploratory, with failed overall reports and unsupported provenance. The full-history candidate passes population, source-preservation, normalization, and reconciliation checks. These establish implementation consistency, not reconstruction accuracy. A separate season holdout of 1,000 games demonstrates the roughness of context imputation: attendance error averages 10,227 people and start-time error 234 clock minutes. Recorded values, estimates, method labels, uncertainty summaries, and unresolved conflicts remain distinct. The result is a reviewable imputation resource and a framework for stating what its evidence supports, rather than a claim to recover the unrecorded game as fact.

1. Introduction

A historical play-by-play account can identify the batter, the play result, and the change in base-out state while omitting the pitches, the ball's trajectory, or the fielder who made an out. These omissions limit comparisons across eras: a statistic computed only where detail survives describes a selected part of the archive. An estimate can make the remaining record usable, but its meaning depends on why the detail is missing and on the assumptions used to fill it.

This paper studies 18,141,020 events in 205,886 acquired play-by-play (PBP) games from 1903 through 2025. Of these, 119,335 games and 10,308,542 events precede 1989. The earliest decade contributes only 41 games; the calendar span is not a claim of complete historical acquisition. We target fields on actual PBP rows. Reconstruction from box scores or season totals is a separate problem because it requires inventing the event spine as well as its missing details.

Two coupled processes produce the archive. The game generates a trajectory, location, sequence of pitches, and sequence of fielding actions. Scorers, sources, translators, and parsers determine which aspects become recorded data. A source may omit a label precisely when another part of the account makes it redundant. In the historical geometry dataset, the pre-1950 unrecorded slice contains 763,993 ground balls deducible from fielding strings, compared with 712,469 explicitly recorded trajectories of all classes. The derived cases impose a lower bound on the unrecorded ground-ball share; they do not make that slice a random sample or identify its

remaining composition. Section 5 develops this argument, and Section 7 identifies the dated evidence behind the counts.

The work has two connected parts. The first is a family of hierarchical coverage models for observation propensity, geometry, fielding credit, handlers, park factors, run expectancy, transitions, and pitch summaries. These models make assumptions and uncertainty explicit, but the available experiments do not establish historical identification or calibrated reconstruction across eras. The second is an additive imputation layer that covers the acquired PBP schema using those retained estimates where available and declared rule-based, empirical, transported, or neutral fallbacks elsewhere. This second part accepts rough estimates while requiring their estimated status to remain visible.

The contributions are a taxonomy at event-dimension grain; a selection-model sensitivity formulation that separates lower bounds from point identification; a full-history PBP field registry and imputation implementation; and an evidence contract that separates numerical convergence, predictive performance, calibration, transport, identification, and provenance. The results include failed and unsupported evidence. Passing a conservation check, or successfully materializing a database table, does not turn an assumption into an observation.

The manuscript’s evidence cutoff is September 13, 2026. Earlier fit summaries are retained as dated research observations. The new PBP surfaces and migrated legacy wrappers have been exercised in isolated database copies; production promotion and public publication of this candidate remain pending. Throughout, “imputed” describes estimated data, including rough reconstructions, and “observed” is reserved for recorded source evidence.

2. The record and its gaps

The database combines source families recorded at different grains. PBP carries event and event-player detail; box scores carry official game-player and game-team totals; gamelogs establish game occurrence and outcomes; season supplements carry season aggregates. The project builds its model layer through SQLMesh over source Parquet files. An official total, a deterministic deduction, an expected credit, and a sampled sequence remain distinct quantities.

The current source snapshot contains 205,886 PBP games and 18,141,020 events spanning 1903-2025. Before 1910 there are 41 games and 3,262 events. The historical manuscript’s 205,845-game count covered 1910 onward while its event count included the early games; this revision uses the same full PBP population for both counts. Games represented only by box, gamelog, or season records receive no generated event rows. Available box totals may constrain attribution within an existing PBP game without bringing box-only games into the target population.

The field registry classifies 179 columns across ten source relations: 116 imputation targets and 63 book-keeping or derived duplicates. In the baseline, 28 targets contain null or unspecified values; accounting for whole missing pitch blocks raises that count to 35. A target can be complete in the source and still belong in the registry because preserving its observed value is part of the output contract. Coverage is reconciled by era, league, source type, and game type, rather than inferred from a single season-level flag.

A missing value does not have one meaning. An absent pitch block differs from an unknown token within an otherwise recorded sequence. A fielding event may have known total outs but unknown player attribution. A missing secondary umpire identity may mean the role did not exist or was not recorded. A contact-strength code of `Default` is unspecified or neutral; treating it as an unobserved `Hard` or `Soft` category would impose a false taxonomy. The imputation layer preserves these distinctions in source values and method or disposition fields.

Geometry requires particular care. Earlier artifacts called six within-zone angle modifiers `location_side`; they do not encode global field side. The legacy consumer now exposes that vocabulary as `location_angle`. The full PBP candidate derives global side, depth, and edge jointly from general location, with separate treatment of angle modifiers. Counts and performance scores for the old target cannot be read as evidence for global-side reconstruction.

The older modeling datasets contain two unrelated game partitions. Deep supplements use DuckDB `HASH(game_id) % 100`: buckets 0-69 for training, 70-84 for validation, and 85-99 for testing. Bayesian fits and coverage gates hold out fold 0 of a ten-fold BLAKE2s hash, removed before subsampling. The deep pipeline also uses five-fold BLAKE2s out-of-fold predictions within its training set. These are different boundaries, not a common outer holdout. Section 6 explains why cross-fitting within one stage is insufficient to certify the whole composed predictor as free of leakage.

Population and registry evidence is retained in the September 13 release manifest, grouped coverage report, and completion registry. The evidence ledger accompanying this paper binds these reports to their file hashes and separates them from the older geometry dataset and its descriptive tables.

3. A taxonomy of missingness

Missingness assumptions must specify a population and conditioning set. Assigning a single field-wide label of missing completely at random, missing at random, or missing not at random is inadequate for this record. A single event can be complete for basic batting, treated under an MAR working model for trajectory given scorer and result, exposed to an MNAR risk for hit location, structurally absent for pitch sequence, and merely aggregate-only for fielding credit, all at the same time. The unit that carries a mechanism is the event-dimension, not the field and not the row, and the mechanism is a statement about a process, not a flag.

Two coupled generative processes produce every value. The baseball process generates the latent state: a ball has a latent contact class and geometry, fielders handle it, runners advance, runs score, and an official scorer assigns credit. The observation process generates the record: a source, scorer, inputter, translator, and parser record or omit parts of that state. Written as a schema, the latent baseball state maps to official credit and outcome, and the latent state together with the scorer-and-source process maps to the recorded label or its absence. Most heuristics blur these two maps; separating them is the point of the taxonomy.

Under that separation, ten missingness classes recur across the record, each with its own detector and its own defensible imputation family.

Class	What it means
Structural absence	The target grain does not exist though a coarser one does - no event rows for a gamelog-only game.
Source-family block absence	The grain exists for some games but a whole source family or file block was never acquired.
Aggregate-only coverage	An official or parser aggregate exists but event attribution does not.
Field-level unknown	The event exists but a parsed field is Unknown, 0, Default, or null.
Selection-biased detail	Detail is recorded only for non-random subsets, keyed to result salience or scorer habit.
Cross-source disagreement	Event-derived and box-derived totals conflict.
State reconstruction gap	The event exists but the state needed to interpret it is derived indirectly.
Official scoring convention gap	The raw facts are known but official credit follows a scoring convention, not event logic.
Taxonomy collapse	Raw codes exist but the wanted category is a coarser, more stable recode.
Sparse-context estimation	Detail exists but the conditioning bucket for adjustment is thin.

These classes call for different responses, and conflating them destroys the model. Structural absence and aggregate-only coverage are not imputation targets in the event namespace at all; they stay at aggregate grain with source flags. Taxonomy collapse is deterministic recode, not inference. Cross-source disagreement requires an explicit authority rule or a retained conflict, rather than a fill that silently discards contrary evidence. Field-level unknowns take deterministic inference first, empirical priors second, and model predictions last, always preserving the raw and imputed values separately. The sentinels themselves must stay distinct: null, Unknown, Default, 0, not-applicable, aggregate-only, and known-source-issue are different markers, and flattening them into one missing indicator discards the missingness model .

Every imputed value therefore carries companion facts for source, method, observed status, and confidence or weight; without them a downstream aggregation cannot tell an official total from a deterministic derivation from a posterior draw from a block-missing gap .

The default working assumption for the observation models is missingness at random conditional on context - source, scorer, inputter, translator, result, hit-or-out state, leverage, era, and base-out state . Two dimensions are flagged as risks to that assumption. Hit location and detailed contact type are treated as missing-not-at-random risks, because whether the label was recorded may depend on what the label would have been . These are the selection-biased-detail class in its sharpest form, and they are handled not by an MAR fill but by pattern-mixture sensitivity - letting the missing values take distributions shifted from the observed ones within bounded plausibility - the subject of the section on selection that never recorded itself. A hierarchical model cannot rescue an unidentified estimand: where scorer, park, team, source, and era are inseparable in a slice, the output is tagged weakly identified, withheld, or supplied only as an explicit assumption-based proxy .

4. A family of coverage models

Two modeling layers coexist in the September 13, 2026 repository, and their evidence must not be combined. The first is the hierarchical Bayesian and deep learning research developed through September 4. Its retained artifacts preserve useful specifications and development results, but all 24 isolated legacy pointers now have gate-version-3 status failed; all have unsupported provenance, identification, and transport evidence. Explicit exploratory pointer authorization allows those bytes to be materialized for research and compatibility testing. It does not turn them into scientifically validated estimates.

The second layer is the September 13 full-history PBP imputation candidate. It completes the acquired play-by-play spine with recorded values, deterministic derivations, normalized empirical distributions, constrained allocations, and broad fallbacks. The candidate covers 205,886 acquired PBP games and 18,141,020 events from 1903 through 2025, and its registry maps 116 completion targets into sixteen additive `pbp_imputed_*` surfaces. It passed artifact-integrity, population, grouped-coverage, schema, and conservation checks in an isolated consumer schema. Those checks establish complete and reproducible coverage of the declared PBP population. They do not establish historical predictive accuracy or convert an estimated field into historical ground truth.

The historical model family

The earlier research organized its estimands as a lettered family. The table is retained as a methods inventory, with current evidence status rather than its September 4 publication status.

Letter	Estimand	Legacy consumer or disposition	September 13 status
A	$P(\text{dimension recorded})$ per event	<code>scorer_observation_propensities</code>	exploratory wrapper; failed gate v3
B	scorer contact-label confusion Ω	withheld	data uninformative (§8)
C	fielding credit share per position	<code>imputed_fielding_credit</code> , <code>assist_count_distribution</code>	exploratory wrapper; failed gate v3
D	handling fielder position	<code>imputed_ball_handler_probabilities</code>	exploratory wrapper; failed gate v3
E	batted-ball geometry class	<code>imputed_batted_ball_geometry</code>	exploratory wrapper; failed gate v3
F	park run effect per season-league	<code>park_factor_summary</code>	exploratory wrapper; failed gate v3
G	run expectancy, base-out transition, run values	three legacy consumers	exploratory wrapper; failed gate v3
H	runner advancement class	typed empty legacy surface	no retained fit
I	fielder responsibility	withheld	required source input absent (§8)
J	final-count coverage and distribution	coverage typed empty; summary wrapped	summary failed gate v3
K	shift propensity	withheld	designed, not fit (§8)

The 19 Bayesian pointers lack their declared `inference/prior_predictive.nc` files. Their current reports can preserve old numerical and limited predictive results, but the missing files and incomplete dependency lineage block provenance. No retained wrapper passes all six gate-v3 dimensions: numerical, predictive, calibration, transport, identification, and provenance.

Shared likelihoods and pooling

The old family used three main likelihood shapes. Recording indicators were Bernoulli logistic models,

$$R_i \sim \text{Bernoulli}(p_i), \quad \text{logit } p_i = \eta_i.$$

Categorical quantities such as contact geometry, handler position, credited fielder, and end state used reference-class softmax models,

$$Y_i \sim \text{Multinomial}(n_i, \pi_i), \quad \pi_{i,c} = \text{softmax}_c(\eta_{i,c}).$$

Count and rate quantities used negative-binomial likelihoods,

$$y_i \sim \text{NB}(\lambda_i, \phi), \quad \log \lambda_i = \eta_i.$$

Group effects were expressed through non-centered partial pooling, $\beta = \sigma z$, $z \sim N(0, 1)$, and $\sigma \sim \text{HalfNormal}(s)$. These equations remain the implemented research specifications. Their presence does not supply the missing observation model, historical transport evidence, or artifact provenance.

Where a geometry model consumed a deep proposal, the proposal entered the class-specific logit as

$$\eta_{i,c} = \alpha_c + \sum_j \beta_c^{(j)}[x_i] + \gamma \log \tilde{p}_{i,c}^{dl}, \quad \gamma \sim N(0, 0.5).$$

The paired `gamma_dl_zero` and `gamma_dl_shrunk` flavors were intended to measure whether that learned covariate materially changed the Bayesian surface. The old ablation outputs needed to reproduce the strongest cell-shift claims are not retained, so those claims are not evidence in this revision. Section 6 records what remains verifiable.

Observation and selection

Model A estimated $P(R_i = 1 \mid x_i)$ using season, scorer, park, and event context on the logit scale,

$$\text{logit } p_i = \alpha + \beta_{t_i}^{\text{season}} + \beta_{c_i}^{\text{scorer}} + \beta_{k_i}^{\text{park}} + \sum_j X_i^{(j)} \beta^{(j)}.$$

Whole games, rather than individual events, were assigned to the Bayesian development holdout. Historical reports found high discrimination for whether a field was recorded, but discrimination of R_i does not identify the missing class distribution $P(Y_i \mid R_i = 0)$. Gate v3 therefore treats the old observation-propensity numbers as dated development evidence, not proof that inverse-propensity or missing-at-random imputation is valid. Section 5 states the selection problem directly.

Geometry and handler research

Models D and E fit class distributions on observed rows and scored unrecorded rows. Their common form was

$$\pi_{i,c} = \text{softmax}_c \left(\alpha_c + \sum_j \beta_c^{(j)}[x_i] + \gamma \log \tilde{p}_{i,c}^{dl} \right).$$

The strongest surviving lesson is semantic. The legacy six-class field called `location_side` contains within-zone angle modifiers, including `Default`; it is not a six-class global field-side target. The compatibility wrapper therefore exposes it as `location_angle`. The September 13 PBP geometry candidate separately derives global side from general location and preserves angle, depth, edge, trajectory, handler, and their method fields as distinct quantities.

The PBP geometry completion uses fixed full-history donor populations and normalized empirical distributions with explicit fallback levels. This supplies an estimate wherever the declared PBP target is applicable. It is a coverage policy chosen for the full-history objective, not a new fit of the old Bayesian geometry model and not evidence that its selected class is historically correct.

Airborne subtype standardization is a separate measurement problem. Recorded Ground versus Air and bunt status are preserved. Fly, LineDrive, and PopUp are translated to standardized airborne bands by season, recorded subtype, and result family. The 2009–2019 translation had useful leave-one-season-out development evidence; the 2020-onward screen failed because a two-season pipeline could not identify its concentration under the declared holdout; pre-2009 values depend on explicit modern-mix and vocabulary-transport assumptions. The accepted estimate is therefore labelled exploratory, with every pre-2009 row partially and weakly identified. The later full-history builder extends that nearest translation to pre-1989 rows as a weakly identified fallback. These outputs extend coverage under explicit assumptions. They do not reopen or reverse the failed scientific screens.

Fielding credit and aggregate constraints

The historical Model C combined a supervised event arm with aggregate box constraints. In its stated form,

$$Y_e \sim \text{Multinomial}(U_e, \pi_e), \quad T_m \sim N\left(\sum_{e \in m} U_e \pi_{e,k}, \sigma_{\text{box}}\right).$$

The supervised arm is necessary because a box total alone cannot determine which event or eligible fielder receives the credit. The September 13 completion layer implements the same distinction without claiming a newly validated posterior: it preserves known official credits, constructs eligible-player candidates, and accepts an allocation only when official residual capacity and event demand have an exact compatible integer assignment. Contradictory or incomplete cases retain an explicit disposition. Conservation certifies the allocation arithmetic, not the identity of an otherwise unobserved historical fielder.

Park, run expectancy, transitions, and pitches

The legacy park model used a team-game negative binomial with plate-appearance exposure, season-league centering, and an AR(1) park history,

$$\theta_{p,t}^{\text{raw}} = \rho^d \theta_{p,t-d}^{\text{raw}} + \varepsilon_{p,t},$$

with a gap-aware stationary bridge for an observed gap of d seasons. The chain still lacks a park-episode reset because the required episode field is absent. The historical run-expectancy model used a negative binomial over base-out cells; the transition and pitch-summary models used structurally masked softmaxes so impossible out decreases and impossible terminal counts carried no free class parameter. These are useful specification corrections retained from the September 4 research. Their wrapper artifacts now fail gate v3, so old convergence and held-out values are reported only as dated development findings.

The full-history PBP candidate reuses existing estimates when available and then backs off to declared season, league, state, park, or corpus donors. Transition fallbacks transport whole normalized vectors; linear weights may use deterministic fallbacks. Pitch reconstruction preserves raw sequences and parser statuses, fills only the declared PBP appearances, and retains source-conflict categories. These operations were checked for row coverage, normalization, and rollup conservation. Their historical accuracy remains unconfirmed.

A scalar random effect under softmax is worth nothing

One algebraic result survives independently of any artifact verdict. A season, scorer, or park effect added as one scalar to every class logit has no effect:

$$\text{softmax}(x + c\mathbf{1}) = \text{softmax}(x).$$

Such a term cannot represent class-specific scorer or era behavior. It only adds an unidentified posterior direction and potential sampling difficulty. A useful group effect in a categorical model must vary by class or enter a different part of the observation process. This correction remains a valid model-design result even though the refitted legacy artifacts do not satisfy the current publication evidence contract.

5. Selection that never recorded itself

The September 4 legacy geometry models learned $P(c \mid x, R_i = 1)$ from events whose class was recorded and applied that distribution to events with $R_i = 0$. Equality of the two conditional distributions is a missing-at-random assumption. The acquired record cannot test it directly because the class is absent on the target rows. Moreover, the chance that a scorer records trajectory plausibly depends on event salience, source practice, and the contact label itself. High held-out accuracy within the recorded slice therefore does not establish accuracy on naturally unrecorded events.

Some unrecorded trajectory rows have fielding descriptions from which the project rules derive a broad contact class. For example, certain shortstop-to-first plays are classified as ground balls under the deterministic deduction rules. These are rule-conditioned derivations, not independent physical measurements and not proof that every superficially similar play was a ground ball. In the September 4 ledger, derived-ground rows numbered 763,993 of 2,615,579 unrecorded pre-1950 events, 1,057,776 of 3,117,160 in 1950–1987, and 53,922 of 134,970 from 1988 on. The counts show that the derivation is consequential; because eligibility for deduction depends on the recorded fielding description, they do not by themselves identify the class mix of the remaining unknown rows.

The selection model

Let a latent class be drawn from $P(c \mid x_i)$ and then recorded with probability $s(c_i, x_i)$:

$$c_i \sim P(c \mid x_i), \quad R_i \sim \text{Bernoulli} \{s(c_i, x_i)\}.$$

Both observed and unobserved slices are products of that selection process:

$$P(c \mid x, R_i = 1) \propto P(c \mid x)s(c, x),$$

$$P(c \mid x, R_i = 0) \propto P(c \mid x)\{1 - s(c, x)\}.$$

For the classes under consideration, assume $0 < s(c, x) < 1$, so both selection outcomes have positive support. Consequently,

$$P(c \mid x, R_i = 0) \propto P(c \mid x, R_i = 1) \frac{1 - s(c, x)}{s(c, x)}.$$

Suppose, as a sensitivity assumption, that the nonselection-to-selection log odds separate as

$$\log \frac{1 - s(c, x)}{s(c, x)} = a(x) + \delta_c.$$

Here $a(x)$ is class independent and δ_c is a class-specific constant. The class-independent component adds the same quantity to every class logit and cancels under softmax. The unrecorded distribution can then be represented as

$$P(c \mid x, R_i = 0) = \text{softmax}_c(\eta_{i,c} + \delta_c).$$

This is an assumed sensitivity form. Selection that interacts between class and covariates cannot generally be reduced to one constant offset per class.

Neither $P(c \mid x, R_i = 1)$ nor the marginal propensity $P(R_i = 1 \mid x)$ identifies δ_c . The observed slice was selected too; it identifies the conditional distribution among selected rows, while the missing labels prevent a direct estimate of how class composition changes when $R_i = 0$. A former design placed a learned class coefficient on Model A’s marginal observation propensity. The September 4 masked experiment reported essentially no reduction in its focal share error. That result is consistent with the identification problem: a marginal propensity cannot recover class-dependent selection without additional truth or a specified selection model. Its old smoke artifact is not retained, so the quoted coefficient and sampler diagnostics are historical notes rather than reproducible current evidence.

What the derived slice supports

The previous paper revision set $\delta_{\text{GroundBall}} = \log(p_{\text{derived}}/p_{\text{obs}})$. Because every row admitted by that derivation was labelled GroundBall, $p_{\text{derived}} = 1$, so the supposed anchor reduced to $-\log p_{\text{obs}}$. It was a function of

the observed slice and supplied no new measurement of the unknown rows. The offsets and corrected shares derived from that construction are withdrawn.

If the deduction rule is accepted as correct on its admitted rows, it supplies a conditional lower bound:

$$P(\text{GroundBall} \mid R_i = 0) \geq \frac{n_{\text{derived ground}}}{n_{\text{unrecorded}}}.$$

The September 4 ledger gives floors of 0.292 before 1950, 0.339 in 1950–1987, and 0.400 from 1988 onward. These are floors conditional on the rule’s validity and the ledger’s population definition. They do not prove MNAR, identify a point share, or validate the derivation on rows without independent labels. No analogous deduction supplies a floor for fly balls, line drives, pop-ups, or bunts.

The same derived rows can be used as a diagnostic subset, again conditional on the deduction. The September 4 MAR export assigned mean GroundBall probabilities of 0.320, 0.402, and 0.375 across the three era groups to rows labelled GroundBall by the rule. The gap is evidence that the model did not encode the deduction signal; it is not an unbiased estimate of error on all naturally missing rows, because the diagnostic subset is selected by the availability and content of fielding detail.

Sensitivity rather than identification

The defensible use of δ_c is an assumption grid. The September 4 analysis reweighted event probabilities as one class offset swept $\pm\{0.25, 0.5, 1.0\}$ nats with other offsets fixed at zero. This operation is a closed-form sensitivity calculation, not a fitted correction. Its width was chosen from one synthetic masking exercise and does not bound the real historical selection mechanism.

The old robustness table used 50 draws, 50 tuning steps, and two chains. All four fits failed their convergence threshold, and the underlying run artifacts are not retained. Three masking designs also made oracle marginal reweighting an algebraic identity. The remaining covariate-interaction design reported that a marginal offset removed about half the induced focal-share error. That is a historical mechanism check under a constructed mask, not validation of historical MNAR correction.

Current coverage policy

The legacy MAR geometry pointers are explicitly exploratory and fail gate version 3. In particular, transport and identification are unsupported. The September 13 full-history PBP candidate addresses a different objective: it supplies normalized empirical distributions and broader declared fallbacks so every applicable target on the acquired PBP spine has an estimate. It does not estimate δ_c or claim that missingness is at random.

Those completed rows must therefore be interpreted as assumption-labelled rough reconstructions. Their method, fallback, source status, and uncertainty fields are part of the estimand. Conservation and complete row coverage show that the outputs cohere with their declared rules; they do not establish the unobserved historical class. No sealed confirmation labels were opened or rescored for this completion work.

6. Deep-learning supplements

Intended role

The intended architecture uses a deep classifier as a proposal distribution, never as recorded truth:

deep proposal $\rightarrow$ out-of-fold probabilities $\rightarrow$ Bayesian layer $\rightarrow$ estimated surface .

An argmax class is insufficient because it discards uncertainty. In the legacy geometry models the proposal entered as $\gamma \log \tilde{p}_{i,c}^{dl}$, with one scalar γ shared across classes and a $N(0, 0.5)$ prior. Separate `gamma_dl_zero` and

`gamma_dl_shrunk` flavors were meant to show whether the learned proposal added signal beyond the hierarchical context model.

This remains a design, not current publication evidence. The isolated September 13 migration contains four deep geometry pointers and one event-universe pretrain pointer. All five have overall status `failed`. Each geometry proposal has numerical evidence marked passed but predictive and calibration evidence marked failed; transport, identification, and provenance are unsupported. Every evidence dimension for the pretrain pointer is unsupported. Explicit exploratory reasons make the compatibility pointers well formed, but do not validate the models.

Target semantics and missing proposals

The old target named `location_side` was misdescribed. Its six categories are within-zone angle modifiers, including `Default`, and are now exposed by the legacy wrapper as `location_angle`. Development scores for that target cannot be interpreted as global left/center/right field-side reconstruction. In the new PBP geometry interface, global side is derived from general location and remains separate from angle, depth, and edge.

The legacy location proposal specifications also scored observed rows but did not supply predictions on the unrecorded production rows they were intended to help impute. Centering a missing zero vector by the training-class mean then introduced a constant class-specific logit shift. The repair made a missing proposal contribute zero and selected deep-free flavors where no production proposal existed. This is a valid implementation finding. It does not make the repaired legacy artifacts pass the current evidence contract.

What the retained development reports support

The September 4 review reported modest TEST-partition improvements over result-family marginals for trajectory, angle, depth, and edge. It also reported larger held-out gains when the trajectory proposal entered the Bayesian model. Those figures remain dated development observations. The strongest ablation claims from the previous draft depended on zero-flavor artifacts and per-cell shift files that are no longer retained, so their exact log-loss deltas and shares of cells moving more than 0.25 posterior standard deviations are omitted here.

Gate version 3 now requires aligned model and baseline probabilities on identical `event_key + partition` rows, declared class truth, improvement in log loss and Brier score, and classwise calibration evidence. The four migrated deep pointers fail because the retained artifacts do not provide acceptable baseline and calibration evidence under that contract. An older numerical pass or a top-1 accuracy improvement cannot substitute for these missing checks.

Leakage in the legacy stack

Two distinct leakage paths prevent confirmatory interpretation. First, the shared event-universe pretrain used geometry labels from the whole `TRAIN` partition, then downstream fold models warm-started embeddings from that fit. A nominally out-of-fold row could therefore influence its own embedding through pretraining. The deep partition and Bayesian game-hash holdout were unrelated, so this contamination also reached part of the Bayesian held-out evaluation.

Second, a trajectory target configured with `fold_count=1` followed a full-fit fallback while labelling its predictions as out of fold. The later five-fold repair addressed that direct error, but it did not repair the shared-pretraining leak or reconstruct complete lineage for the retained artifacts. The dependency manifest needed to identify exactly which pretrain produced each proposal is also absent.

The scientific cross-fitting contract is stricter than a prediction-scope label. Every supervised stage that can encode the target must be fit inside the outer training games. Predictions must be generated on disjoint games, carry explicit scope and class labels, and be compared with a baseline on the same rows. Any calibration

stage must also be trained without the evaluation rows. The retained legacy stack does not demonstrate that complete contract.

Relation to the September 13 completion layer

The full-history PBP imputation candidate does not require a new deep fit. It uses recorded values, deterministic derivations, empirical donor distributions, constrained reconciliation, and explicit broad fallbacks. Legacy probabilities may be carried as exploratory inputs where configured, but a complete candidate row does not inherit scientific validation from an old pointer. The builder keeps the method and artifact identity visible so consumers can distinguish a recorded fact, a deterministic rule, a transported estimate, and a broad prior.

This separation is deliberate. The coverage objective accepts rough estimates across the acquired historical PBP population. More deep training would answer a different question and would require fresh lineage, leakage-free outer folds, retained baseline predictions, calibration, transport tests, and untouched confirmation data. None of that work was performed for the September 13 candidate.

7. Results and evidence status

The results separate historical model summaries from the September 13 PBP candidate. The earlier tables describe a frozen research generation; the new reports measure coverage and implementation consistency. Neither supplies unobserved historical ground truth. The paper's evidence ledger identifies the reports and their hashes, and retains the older table snapshots with their dates.

Full-history PBP coverage

The selected candidate covers 205,886 games and 18,141,020 events from 1903-2025, including 119,335 games and 10,308,542 events before 1989. Every existing PBP game and event has an output row in its corresponding combined interface. This is complete coverage of the acquired PBP population, not of every major-league game that occurred. The registry maps all 116 targets to their output and evidence columns. Source preservation, explicit non-applicability, and unresolved conflicts are valid dispositions; coverage does not mean every field becomes a known historical value.

Eleven component artifacts feed sixteen database surfaces. Component row counts have different grains and must not be added as if they were distinct events.

Component	Rows	Grain
Game context	205,886	PBP game
Pitches	18,141,020	Event increments within appearances
Event values	18,141,020	PBP event
Geometry	12,041,398	Applicable batted-ball event
Runners	17,411,649	Runner record
Fielding	15,719,352	Fielding-play record
Officials	1,441,202	Game and official role
Park factors	72,930	Park, league, season, and metric
Run expectancy	8,473	PBP base-out context
State transitions	211,825	Start context and end state
Linear weights	7,640	League, season, and play

Source: selected component metadata and the release-candidate manifest, September 13, 2026. The 12,041,398 geometry rows use the full PBP scope and current eligibility definition; the older geometry fit population below is a separate dataset generation.

The combined interfaces are `pbp_imputed_games` and `pbp_imputed_events`. Specialized surfaces expose game context, geometry, pitches, individual pitch items, pitch totals, runners, fielding plays, fielding totals, officials, event values, park factors, run expectancy, state transitions, and linear weights. All use the `main_models.pbp_imputed_*` namespace. They preserve recorded values and make imputation methods available for downstream filtering or aggregation.

The full validator found no missing or duplicate game/event keys, population count differences, or non-PBP additions. All sixteen SQLMesh consumer audits passed, as did materialized artifact-identity checks and the field mapping checks. Coverage by era, league, source, and game type reconciled with the baseline for every target. These are structural guarantees: they do not assess whether a missing value has been recovered accurately.

Reconstruction constraints and disclosed conflicts

Pitch validation covers 15,822,702 appearances across all 18,141,020 events. The independent audit found zero counter discrepancies across seventeen checks, zero unknown tokens, zero unclassified dispositions, and zero unflagged appearance violations. Separately disclosed source violations include 26,996 ball-boundary cases, 51,245 strike-boundary cases, 816 post-terminal cases, and 375 terminal-outcome cases. Categories overlap and are not counts of distinct bad games. A consistent constructed sequence is one possible sequence under the rules and assumptions, not a recovered historical sequence.

Runner end-state identity resolves 634 of 642 missing non-out destinations; eight terminal or frame conflicts remain explicit. Fielding allocation uses known credits and compatible official totals within existing PBP games. An allocation is called successful only when the integer capacities match and the allocation delta is zero. Merely having aggregate inputs is insufficient. Contradictory evidence is retained rather than overwritten to force agreement. Official identities use contemporaneous candidate support when available; unresolved slots do not create fictitious people.

Geometry preserves a joint general-location, global-side, depth, and edge representation. Historical angle modifiers remain separate. Pre-1989 airborne standardization transports the nearest available 1989 translation backward, with a weak-identification flag. The selected geometry repair fills 1,123 early non-bunt standardization gaps; 157,180 bunts remain explicitly outside that standardization target. These additions extend computational coverage under the existing airborne research assumptions, without adding confirmation evidence.

Derived run and win changes, park factors, run expectancy, transitions, and linear weights retain existing nonmissing values or estimates. Their remaining gaps use declared nearby-season, pooled, deterministic, or neutral fallbacks. Whole transition vectors are transported together and normalized. Conditional donor variation and transferred uncertainty summaries do not cover all model, era-transport, or source-selection uncertainty.

Recorded-context season holdout

A separate stress test jointly masks ten context fields in a 1,000-game sample from complete held-out seasons 1903, 1910, 1940, 1960, 1980, 2000, and 2025. Each target's donor pool excludes those seasons. The denominators below count only held-out games with a recorded value for that target. The audit detects no held-out source values reused as donors.

Numeric target	Recorded test values	Mean absolute error
Temperature	543	7.53 degrees Fahrenheit
Attendance	944	10,227 people
Wind speed	403	4.32 mph
Start time	578	234.34 clock minutes
Game duration	999	25.67 minutes

Categorical target	Recorded test values	Exact agreement
Time of day	1,000	48.3%
Sky	541	48.4%
Field condition	78	87.2%
Precipitation	282	94.0%
Wind direction	396	19.7%

Source: recorded-context holdout report, September 13, 2026. Agreement is not adjusted for a majority-class baseline. This intentionally difficult masking exercise removes context often present when one field is missing. The large start-time and attendance errors make the practical limitation concrete: coverage can be useful while individual reconstructions remain very rough. The experiment does not estimate error on the historically missing stratum, validate airborne labels, or calibrate total uncertainty.

Historical trajectory evidence

The earlier geometry dataset separates recorded trajectories from trajectories deduced from fielding strings and genuinely unknown cases. All derived cases in this dataset are ground balls. Its retained table reports:

Era	Unrecorded trajectories	Derived ground balls	Ground-ball floor	MAR probability on derived cases
Before 1950	2,615,579	763,993	0.2921	0.3203
1950-1987	3,117,160	1,057,776	0.3393	0.4020
1988 onward	134,970	53,922	0.3995	0.3751

Source: historical `groundball_mnar` table and frozen geometry dataset used by the September 4 manuscript. The floor divides derived ground balls by all unrecorded trajectories. It is conditional on the correctness of the deduction rules and this target vocabulary. The last column evaluates a missing-at-random (MAR) fit on known ground balls, whose target indicator is one; it is not the unknown stratum's true ground-ball share. The selected nature of the derived slice prevents identifying that share from this comparison alone.

For the pre-1950 slice the old fixed-offset sensitivity table ranges from 0.162 to 0.530 over a plus-or-minus-one-nat grid, with the 0.292 floor inside it. This is an assumption sweep, not a confidence interval or a data-estimated selection offset. Historical masked experiments in Section 5 were smoke-budget diagnostics; their outputs are not a substitute for a retained confirmatory experiment. The supporting tables remain useful for explaining the derivation, but do not earn a present-day validation pass.

Legacy model evidence under the current contract

The retained legacy statistical consumers contain ten populated families and two typed-empty optional surfaces. For example, geometry contains 253,013,169 class-probability rows, fielding credit 8,335,674 rows, and handler probabilities 11,712,096 rows. These counts reflect class-level expansion and differ from the new PBP component grains. The legacy six-class angle vocabulary accounts for 41,938,992 geometry rows, exposed as `location_angle`, with no rows mislabelled as `global location_side` in the isolated migration check.

The current migration creates 24 explicitly exploratory pointers. Every saved version-3 report has overall status `failed` and provenance `unsupported`; numerical and some predictive components can pass within that

failed overall assessment. Missing files, incomplete dependencies, unsupported calibration, and transport or identification gaps are not repaired by wrapping retained payloads in new manifests. The isolated consumer rehearsal passes all twelve SQLMesh consumer audits and 56 identity, schema, count, and probability checks. Those results establish correct consumption of the selected exploratory artifacts, not statistical acceptance.

The September 4 version-2 table reported 19 passing registered gate targets and four unresolved deep-pointer targets. It also reported observation-propensity ECE from 0.0049 to 0.0181, and interval diagnostics of 0.9780 for transitions and 0.9062 for run expectancy. These are dated diagnostics under the older contract. In particular, the run-expectancy interval uses held-out sample variance and a Normal mean approximation; it is now labelled `studentized_mean`, not a fitted negative-binomial posterior-predictive check. The transition calculation approximates marginal uncertainty without a full joint posterior. Neither establishes historical calibration or joint uncertainty for derived run values.

Later development comparisons reinforce that caution. The shared-split geometry reference fits passed numerical diagnostics but failed their frozen predictive criteria; trajectory also regressed on the pre-1988 observed slice. The old location target was subsequently identified as angle modifiers, making its scores unsuitable evidence for global-side reconstruction. These findings support retaining explicit baselines and assumption labels. They do not support the earlier manuscript’s blanket description of the surfaces as validated or calibrated across baseball history.

8. What the record cannot tell you

Complete output coverage and statistical identification are different goals. The September 13 candidate supplies a reproducible estimate or declared fallback for applicable targets on the acquired PBP spine. That policy does not make an unobserved historical fact learnable from a single source. Three legacy model letters illustrate three different limits.

Model B - contact-label confusion

Model B proposed a latent contact class Z_i and a scorer or era confusion matrix Ω generating the recorded label L_i :

$$Z_i \sim P(Z \mid x_i), \quad L_i \sim \Omega_{g_i}[Z_i, \cdot].$$

Estimating Ω requires repeated independent labels for the same event or a second measurement that can disagree with L_i without being constructed from it. The acquired PBP surface usually provides one recorded contact label. The project’s deduced broad class is not a general second measurement: on many recorded rows its rule uses or reproduces the recorded class, while on other rows it depends on fielding outcomes that are themselves selectively recorded.

The September 4 audit found only a thin outcome-anchored subset with information independent enough to challenge the recorded label. That subset can move a simple aggregate confusion rate, but the retained evidence does not support a scorer-by-era matrix or historical transport. It is therefore more precise to say that this record is uninformative for the proposed structured Ω than to claim a formal non-identification theorem or that scorer confusion is rare. Unblocking the estimand requires an independently generated second contact label at event grain, with its own provenance and observation model.

Model I - analytical fielder responsibility

Fielder responsibility asks who should have had a play given geometry and alignment:

$$P(\text{responsible} = k \mid \text{geometry, alignment, } \dots).$$

That differs from the handling fielder, who touched or retrieved the ball. Training on `ball_handler_position` would simply reproduce Model D with a narrower vocabulary and would be especially misleading on hits, where the retriever need not be the defender with the opportunity.

A defensible model would need a zone or positioning kernel such as $\pi(k \mid \text{location, alignment})$. The current source does not contain tracking-era defensive positions or an independent opportunity label from which to estimate that kernel. This is a data-availability limit, not a claim that responsibility is unidentifiable in principle. The September 13 fielding candidate can allocate official residual credits when eligibility and integer box capacity agree, but official credit reconciliation does not identify analytical responsibility.

Model K - shift propensity

Model K was scoped as

$$P(\text{shift} \mid \text{player, batter hand, defending team, count, outs, base state}).$$

It was designed but not fit. No retained result supports a claim about its identification or accuracy. Geometry and responsibility work that would consume it instead uses coarse alignment regimes or broader fallbacks with explicit method labels. K is unfinished research scope, rather than an identified failure of the source.

Publication consequence

These cases share a reporting rule, not a statistical proof. A structured confusion model without an independent label, a responsibility model without a positioning source, and an unbuilt shift model cannot become validated facts through a complete SQL table. A rough fallback may still be useful for an authorized coverage interface if it is labelled as an assumption, retains its method and provenance, and is kept separate from recorded facts. It must not be described as if the historical record identified the latent quantity.

9. Publication policy

Publication status and evidential support are separate properties. The model registry distinguishes official values, deterministic derivations, estimates, synthetic data, and withheld outputs. The current PBP layer uses the explicit `pbp_imputed_*` names even where a view combines recorded values with estimates. The source columns remain available, and estimates carry their method, artifact identity, and applicable uncertainty or conflict fields. These names make estimated status visible at the point of use.

Legacy probabilistic surfaces carry eight common provenance fields: `artifact_id`, `model_name`, `model_version`, `source_snapshot_id`, `method`, `observed_status`, `confidence_status`, and `weak_identification_flag`. A posterior mean or HDI is conditional on the fitted model. Donor dispersion, a transported interval, or a neutral fallback in the PBP layer is a different uncertainty summary and must retain its own method label. The PBP layer is not uniformly Bayesian and does not promise calibrated intervals for every target.

The September 11 evidence contract, gate version 3, reports six dimensions: numerical, predictive, calibration, transport, identification, and provenance. Each dimension can pass, fail, or remain unsupported. Default validated publication requires current numerical, predictive, calibration, and provenance passes. Even that status leaves transport and identification as separate judgments. Finite metrics and positive evaluated sample counts are required; a missing comparator, missing calibration evidence, or incomplete fitted-stage lineage cannot be treated as a pass.

An explicit exploratory publication mode permits a research pointer with failed or unsupported evidence and a recorded reason. It preserves the failed result. The September 13 migration creates 24 such pointers bound to retained legacy payloads; all have failed overall reports and unsupported provenance. Nineteen missing prior-predictive files and other lineage gaps remain disclosed. A strict pointer check passing means

that this exploratory status and its content binding are internally consistent. It does not mean the model passed validation.

The older paper reported version-2 passed labels after the September 4 restatement. Those labels describe a historical materialization under an older policy. Updating validation code does not revise rows already stored in production. The isolated candidate’s ten populated legacy statistical families carry failed confidence; two optional statistical surfaces remain typed empty. This revision therefore reports the policy version, artifact generation, and consumer environment whenever it reports a verdict.

For the full PBP candidate, publication checks enforce exact population coverage, field mappings, artifact identities, normalization, and pitch and fielding reconciliation. All sixteen PBP surfaces passed their structural audits in an isolated candidate schema. Explicit conflicts remain in the data; checks require their disclosure rather than erasing them. The publisher repeats the relevant checks against production consumers before creating the public catalog. No production promotion or public upload of this candidate had occurred at the paper’s cutoff.

Official tables retain their meaning. Probabilistic legacy outputs remain additive siblings of deterministic tables, and the rough PBP reconstructions are another explicit interface. Unknown people are represented by candidate support or unresolved slots, never fabricated person identifiers. Unidentified physical or scoring quantities may be withheld, while practical proxies can be supplied under declared assumptions. Neither a selected class nor a narrow conditional interval is permission to present an unrecorded value as an observed fact.

10. Related work

The framing rests on the missing-data literature. Rubin (1976) established the distinction between missing completely at random, missing at random, and missing not at random, and the conditions under which the missingness mechanism can be ignored for likelihood and Bayesian inference. Our observation models are a direct application: the default assumption is missingness at random conditional on scorer, source, result, and era, and the contribution is to make that conditioning set explicit rather than assumed. Little and Rubin (2019) give the methodology this paper follows for handling incomplete data through models rather than deletion or single imputation.

Where ignorability fails, two traditions divide the territory, and this paper draws on both. The first is the selection-model tradition. Heckman (1976, 1979) formalized sample selection as a two-equation system - an outcome process and a selection process whose errors are correlated - and showed that selection bias is a specification error correctable by modeling the selection equation jointly with the outcome. §5 uses a categorical selection formulation: a latent class is drawn, then recorded with probability $s(c, x)$, and Bayes’ rule turns the selection process into a per-class reweight of the fitted distribution - the offset δ_c is the per-class selection log-odds, a selection-model parameter acting on the recorded class before the softmax normalizes. The difference from Heckman’s program is what we ask of the parameter. Heckman buys identification with parametric structure - joint normality of the two equations’ errors, or an exclusion restriction that moves selection without moving the outcome. We have not established a credible exclusion restriction in this study. Scorer, source, and era covariates can also proxy the game process. We therefore treat δ_c as unidentified from the recorded labels alone and report sensitivity to assumed values. A partial-truth slice constrains the missing ground-ball share from below; it does not independently identify δ_c .

That sweep places the paper in the second tradition: pattern-mixture models and delta-adjustment sensitivity analysis. Little (1993) specified the distribution of missing values directly through identifying restrictions rather than a mechanism, and observed that such models are chronically underidentified - without additional

restrictions the observed data do not identify the missing stratum's distribution. Inference across alternative restrictions therefore makes the unresolved assumption visible. Scharfstein, Rotnitzky, and Robins (1999) made the same move inside the selection formulation: fix a nonidentified selection-bias parameter at each value in a plausible range, estimate under each, and report the trajectory - sensitivity analysis in place of an untestable assumption. That template became standard practice in the clinical-trials missing-data literature as delta adjustment and tipping-point analysis, where a fixed offset δ is added to values imputed under a missing-at-random fit and swept until the conclusion changes; Molenberghs and Kenward (2007) treat the framework at book length, and Cro, Morris, Kenward, and Carpenter (2020) give a practical guide to controlled multiple imputation in this style. The sensitivity analysis of §5 is this device transplanted: δ_c is a delta adjustment on the class logits, the missing-at-random fit is the $\delta = 0$ point, and the ribbon is the tipping-point trajectory, reported per class. What the paper adds to the template is a hard constraint on the sweep - a deduced-trajectory partial-truth slice internal to the record that places a floor under one class's share on the missing stratum, and a known-truth subslice on which the missing-at-random fit can be scored - and historical smoke-budget masking diagnostics that illustrate the offset's behavior under a selection process it does not nest. An earlier revision read the same slice as an estimate of δ_c ; §5 explains why it is a floor and not an estimate. The structure of the treatment - a selection-model parameterization with pattern-mixture-style sensitivity reporting - reflects the standard observation that the two factorizations describe the same joint distribution and can be mixed rather than chosen between.

The observation process studied here is informative nonresponse in the survey statistician's sense: the probability an item is recorded depends on the value the item would have taken. Rubin (1977) formalized subjective assumptions about nonrespondents in exactly the spirit adopted here - a parameter the data cannot estimate, elicited and varied rather than ignored. The survey-methodology literature collected in Groves, Dillman, Eltinge, and Little (2002) treats nonresponse as a process with its own covariates and propensities, which is Model A's stance toward scorers: the observation-propensity surfaces are response-propensity models with scorers in the role of respondents, and the event-dimension grain of §3's taxonomy is item nonresponse rather than unit nonresponse - one event can respond on batting and not on trajectory.

The fitting methodology follows the applied Bayesian workflow. Gelman et al. (2013) is the reference for the hierarchical models, partial pooling, prior and posterior predictive checking, and the practice of treating a model as a hypothesis to be checked against held-out data. Betancourt and Girolami (2013) document the pathologies that hierarchical models present to Hamiltonian Monte Carlo and explain why a non-centered parameterization can improve sampling when data are sparse - the regime of most of our early-era cells - which is why every model in the family is written non-centered.

In sports analytics, the play-by-play data itself comes from the Retrosheet archive. The closest methodological predecessor is openWAR (Baumer, Jensen, and Matthews, 2015), an open, reproducible player-value system that propagates uncertainty rather than reporting only point values. Marchi and Albert (2014) is an applied introduction to baseball data analysis. Measurement choices in the baseball record have also been studied directly: Kalist and Spurr (2006) find evidence of home-team bias in official scorers' hit-versus-error calls. That result motivates treating scorer behavior as part of the observation process, without presuming that a scorer decision can be retrospectively corrected. Acharya et al. (2008) show that conventional park-factor formulas can be distorted by unbalanced schedules and inflationary bias, and propose an ANOVA weighted fixed-effects estimator. It is relevant as an example of a baseball quantity whose estimate depends on an explicit model for the recording and game context, rather than as evidence for this paper's hierarchical model.

Our target differs from these studies: we characterize the observation process for event details and report uncertainty in its fitted coverage surfaces. The full-history play-by-play output remains an imputed candidate,

not a validated historical reconstruction; the paper does not claim to recover the unrecorded details as ground truth.

11. Limitations and open problems

The principal limitation is evidentiary. The September 13 work completed and reconciled a full-history PBP candidate, while the retained September 4 model artifacts fail the current scientific publication contract. These statements can both be true because coverage validation and model validation answer different questions.

The legacy statistical family does not pass gate version 3. All 24 migrated pointers have overall status `failed`; provenance, identification, and transport are unsupported for every pointer. The 19 Bayesian artifacts lack their declared `prior_predictive.nc` files. The four geometry deep proposals fail current predictive and calibration evidence, and the event-universe pretrain is unsupported on every evidence dimension. The strict pointer checker passes the exploratory wrappers because their identities, reasons, failed verdicts, and retained-byte bindings are internally consistent. That is an integrity result, not scientific authorization.

The full-history candidate is a rough reconstruction of the acquired PBP population. It covers 205,886 games and 18,141,020 events from 1903–2025 and maps all 116 declared completion targets. Its isolated checks establish artifact identity, source preservation, row coverage, schema compatibility, probability normalization, and defined conservation rules. They do not show that an unrecorded temperature, pitch, fielder, trajectory, official, or run value was historically correct. Games without an acquired PBP event spine, including aggregate-only games, remain outside the estimand.

Missingness remains unidentified for the naturally unrecorded class. A model fit on recorded labels estimates $P(Y \mid X, R = 1)$, not $P(Y \mid X, R = 0)$. Marginal observation propensity does not recover the class-dependent selection odds. Rule-derived ground-ball rows provide at most a conditional lower bound and a selected diagnostic subset; they do not prove MNAR or identify the other trajectory classes. The September 13 candidate uses broad empirical fallbacks rather than claiming a fitted MNAR correction.

The location target correction changes the meaning of old results. The legacy six-class `location_side` artifact contains within-zone angle modifiers and is now exposed as `location_angle`. Its historical predictive scores do not validate global field-side reconstruction. The new PBP interface derives global side from general location and keeps angle, depth, and edge distinct, but the fallback accuracy of those historical estimates is not confirmed.

Airborne standardization is assumption labelled. The 2009–2019 pipeline has useful development support. The 2020-onward held-out screen failed under its two-season design, and pre-2009 translation assumes a modern true-band mix and a relationship to the 2020-onward vocabulary. Its nominal intervals omit uncertainty in those assumptions. The PBP candidate carries the nearest 1989 translation backward for earlier seasons with weak-identification flags. This coverage decision was explicitly accepted as an estimate; it is not a passed transport result.

The confirmation reserves remain sealed. The 121 deferred modern-angle games and 6,105 historical component-local reserve games were not opened for this paper revision or the PBP candidate. No new model was fit and no new confirmation data were acquired. The air-research decision remains closed under its stated assumptions; the later failed screens and partial-identification findings are reported without reopening the reserve.

Legacy deep evidence is incomplete and partly contaminated. Shared pretraining could encode a downstream row’s geometry label before nominal cross-fitting, and the Bayesian and deep partitions were not nested. An earlier one-fold trajectory configuration also labelled full-fit predictions as out of fold. The

strongest ablation files quoted by an older draft are not retained, so their exact effects cannot be reproduced. A future deep supplement would need target-free outer folds across every supervised stage, explicit dependency lineage, aligned baseline probabilities, calibration, and transport tests before it could support a scientific claim.

Reported uncertainty is conditional on method and fallback. Donor dispersion measures empirical variation within the selected donor pool. It does not include uncertainty that the fallback hierarchy, historical transport assumption, or source measurement is wrong. Pre-2009 airborne intervals omit the error in their two main transport assumptions. Legacy run-expectancy and transition interval diagnostics use approximations rather than retained joint posterior draws, and park factors lack a held-out coverage hook. A narrow conditional interval can therefore coexist with weak historical identification.

Conservation does not resolve source conflict. Pitch and fielding reconciliation ensures that the candidate's defined counters and allocations add up. The builder retains parser statuses, raw sequences, and explicit conflict dispositions because many source sequences violate modern or internal boundaries. A reconstructed legal sequence is one coherent completion of the retained evidence, not proof of the historical pitch order. Likewise, exact fielding capacity can constrain a credit allocation without independently identifying the responsible fielder.

The available context stress test shows material roughness. In the retained 1,000-game joint holdout, mean absolute error was 7.53 °F for temperature, 10,227 for attendance, 4.32 mph for wind speed, 25.67 minutes for duration, and 234.34 clock minutes for start time; categorical exact agreement ranged from 19.7% for wind direction to 94.0% for precipitation. This is a test against recorded values under an artificial joint mask, not a calibrated study of historical missingness, but it demonstrates why completed context fields should retain their methods and donor dispersion.

Several latent baseball quantities remain outside the supported source. The record does not supply an independent event-level contact label for a structured scorer confusion matrix or a positioning/opportunity source for analytical fielder responsibility. Shift propensity was designed but not fit. The legacy advancement and pitch-count-coverage consumers remain typed empty because they have no retained validated pointer. The broader PBP interfaces may still expose rough completed values where the registry declares a fallback, but that does not resolve these scientific estimands.

Future scientific promotion would require new artifact IDs, complete retained lineage and prior-predictive outputs, leakage-free grouped holdouts, calibration, source-block and era transport tests, and validation against data not used during development. The current paper instead reports the completed PBP candidate as an exploratory, assumption-labelled analysis layer and dates older model results to the evidence that produced them.

12. Reproducibility and availability

This manuscript is assembled from the ordered Markdown files in `notes/paper/sections/`. Its build command is `node notes/paper/build.mjs` from the repository root. The build uses Pandoc and Typst, removes internal drafting comments, and writes the assembled Markdown and PDF. The accompanying evidence ledger records the source generation, report identities, and limitations of the numbers used in the current results. Historical query files and table snapshots remain under `notes/paper/queries/` and `notes/paper/tables/`; their dates matter.

The current candidate's source database has SHA-256 `fc94ec74507ceb0e7ab60296006de534253d364b0c9d5b64f352c82749c23f9c`. The release manifest selects the PBP root and binds its release evidence. The selected component manifest binds eleven data files, their SQL, frozen implementation dependencies, source identity, and validation reports. Its geometry input is `geometry_v2.parquet`;

earlier geometry outputs remain diagnostic artifacts and are not the selected candidate. The namespace revision changed metadata and output mappings while preserving the eleven component payloads byte for byte. The release code snapshot is commit 98da62c.

Candidate paths and the exact report digests are recorded in the paper’s machine-readable evidence snapshot. The source reports are local research artifacts, not all checked into Git or distributed with the public database. This repository supplies code and an auditable summary; it does not yet supply a self-contained public replication archive. In particular, several older ablation and smoke-backtest run artifacts are not retained, and the legacy migration cannot recover missing prior-predictive files or missing dependency lineage. Quoted historical summaries are documentary evidence, not a claim that every experiment can currently be regenerated from an intact frozen bundle.

The imputation builder reads its source database without modifying it. A small sample is the default; a full build must explicitly request the full population. The commands and artifact-selection environment variables are documented in `docs/pbp-imputation.md`. SQLMesh materialization targets an isolated database and state copy for candidate testing. Verification checks the actual consumer views and artifact identities, because changing an artifact-root environment variable alone need not change a model fingerprint. Neither a successful build nor this paper’s PDF export is a production release.

The recorded-context stress test retains its query, predictions, script, and report. Entire selected seasons are removed from each field’s donor pool before predicting the 1,000 sampled games. The reported error denominators are the numbers with recorded truth for each field, not 1,000 for every metric. This experiment is separate from the older Bayesian and deep partitions described in Section 2, and from sealed geometry confirmation data. The 121 modern-angle games and 6,105 historical reserve games were not opened or rescored for the PBP candidate or this manuscript revision.

The operational database is built through SQLMesh. The current public distribution pipeline copies approved model and seed tables into a DuckLake catalog with immutable Parquet data files, semantic views, and metric macros; the website attaches that catalog read-only. Older standalone database objects are retained for compatibility and rollback. This distribution mechanism does not imply that the September 13 candidate has been released or that its local research artifacts are downloadable from the public catalog.

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